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Agentic AI for Oilfield Optimization: Bridging Expert Workflows and Generative Intelligence in Multi-Field Facilities

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Abstract

This paper presents the deployment of Agentic AI within ADNOC's Artificial Intelligence Production System Optimization (AiPSO), a strategic initiative aimed at transforming upstream oilfield operations through intelligent automation. At its core, AiPSO embeds domain-specific generative agents into engineering and optimization workflows, enabling autonomous diagnostics, scenario modeling, and decision support. These agents interact with a field-wide digital twin powered by hybrid physics/ML models and a knowledge graph that contextualizes data from IT, OT, and ET domains.

The system goes beyond rule-based automation by introducing agents that reason over constraints, simulate outcomes, and proactively recommend actions delivering conversational intelligence grounded in engineering logic and real-time operational data. These capabilities are tightly integrated with foundational workflows including artificial lift diagnostics, MPFM validation, and injection optimization, ensuring adoption and value from day one.

Aimed to be deployed across 25 fields, AiPSO will enable uplift in production capacity with minimal CAPEX while reducing decision latency and enhancing operational transparency. The phased architecture of AiPSO ensures scalability, trust, and explainability key to industrial AI. This paper outlines how Agentic AI transforms traditional workflows into intelligent systems, positioning ADNOC's upstream assets for autonomous operations and redefining how human-machine collaboration evolves in energy production

Introduction

The upstream oil and gas industry is navigating unprecedented challenges: maximizing recovery from mature reservoirs, reducing operational cost, achieving net-zero ambitions, and managing growing data complexity. Digital transformation is no longer a technology experiment but a core operational necessity. Traditional production management tools characterized by dashboards, static reports, and threshold-based alerting—have plateaued in impact. The industry is now demanding systems that are adaptive, autonomous, and explainable. This paradigm shift necessitates a move towards intelligent automation, where Artificial Intelligence plays a pivotal role in optimizing complex workflows and enhancing decision-making processes

across the value chain [1]. The integration of AI, encompassing machine learning and predictive analytics, empowers decision-makers with real-time insights, thereby optimizing production and giving solid, data-based grounds for decision making and opportunity identification.

ADNOC recognized these imperatives and launched AiPSO (Artificial Intelligence Production System Optimization), a strategic transformation framework focused on digitizing field operations with embedded intelligence. AiPSO introduces a shift from digital assistance to decision autonomy, powered by Agentic AI. In this paradigm, generative AI agents are not just query tools—they are collaborators that understand operational objectives, simulate scenarios, evaluate constraints, and deliver engineering-informed recommendations. Agentic AI in AiPSO represents a new generation of upstream digital capability: one that combines deep domain knowledge with reasoning and learning. Unlike earlier automation efforts, these agents interact directly with digital twins, surveillance models, and workflow engines. This enables proactive diagnostics, closed-loop optimization, and intuitive human-AI interfaces for field teams.

This contribution details the conceptual framework, system architecture, implementation outcomes, and strategic ramifications of deploying Agentic AI within AiPSO across ADNOC's upstream operations. Rather than entering in the details of each AI workflows, it illustrates the overall structure of the project and how AI, when contextualized within an engineering framework, can transition from augmenting human capabilities to enabling autonomous decision-making.

The AiPSO program encompasses three distinct phases, with each phase involving the selection of a specific set of fields for the deployment of solutions. Presently, AiPSO is successfully deployed in Phase-1 fields comprising 6 onshore and 2 offshore fields. The overall AiPSO framework is standardized across all 25+ fields of ADNOC. This paper will be focusing on implementation and rollout for the phase-1 fields.

Methodology

ADNOC's production system faced challenges typical of complex, high-volume upstream companies where IT/OT/ET data are siloed and their integration for holistic operational visibility and optimized decision-making remains a significant hurdle [1]. This led to inefficiencies stemming from delayed access to critical information, suboptimal resource allocation, and reactive rather than proactive problem-solving. This scenario underscored the need for an integrated system capable of harmonizing diverse data streams to provide actionable insights for improved operational performance and strategic planning [2].

The integration of AI techniques has been identified as a critical missing piece in addressing various reservoir, drilling, and production challenges within the oil and gas sector [3] to overcome limited real-time diagnostics and high reliance on manual decision-making. Artificial intelligence and machine learning methods, where available, were often fragmented and underutilized, possibly due to a lack of trust and explainability [4]. To overcome these obstacles, AiPSO was conceptualized as a multi-layered intelligent system, designed to integrate diverse data sources in a cohesive data platform, based on an industrial knowledge graph. Then, a series of domain workflows were included, to compute advanced analytics based on contextualised sensor data. These workflows leverage advanced machine learning models anomaly detection, production forecasting, etc. moving beyond traditional statistical methods to incorporate dynamic operational parameters [5]. Every workflow was then augmented with Agentic AI to provide conversational intelligence and automate complex decision-making processes, thereby enhancing efficiency and reducing human intervention. This architectural evolution transforms raw data into actionable intelligence, enabling predictive maintenance, optimized drilling operations, and enhanced reservoir management through autonomous reasoning capabilities [6] [7]. Strategic objectives included improving production efficiency without increasing CAPEX, shortening decision cycles, laying the foundation for unmanned, autonomous operations, and building trust in AI through transparency, traceability, and domain alignment.

ADNOC's Agentic AI-Driven Production System Optimization (AiPSO) Framework

The AiPSO framework represents a strategic initiative aimed at transforming upstream oilfield operations through advanced AI-driven workflows and intelligent automation. Built on a robust technology platform, AiPSO integrates domain-specific generative agents, smart engineering models, and a comprehensive digital twin to deliver autonomous diagnostics, scenario modeling, and real-time decision support.

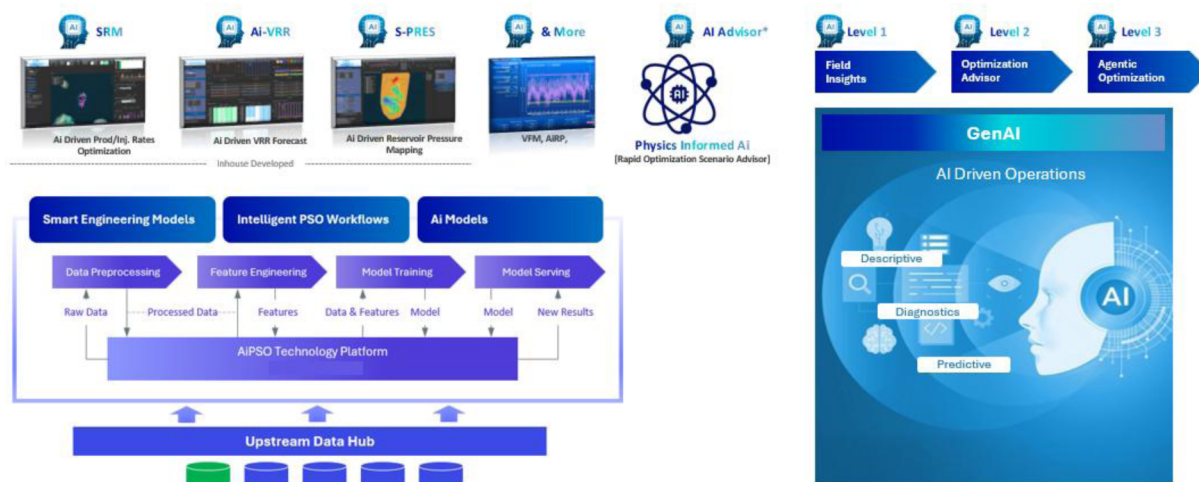


Figure 1—Agentic AI-Driven Production System Optimization (AiPSO) Framework

The architecture is designed around a multi-layered approach:

- Smart Engineering Workflows** – Embedded across all levels, these workflows integrate physics-based models with AI-driven insights to ensure engineering rigor, validate optimization recommendations, and maintain operational integrity. They serve as the bridge between traditional engineering practices and advanced AI capabilities, enabling seamless adoption and trust in AI-driven decision-making.
- Generative Intelligence for Field Operations** - leverages domain-specific generative AI and agentic reasoning to enable autonomous diagnostics, scenario modeling, and real-time decision support. This capability transforms static workflows into adaptive, intelligent systems, driving operational agility and efficiency across upstream assets.
 - Level 1: Field Insights – Provides descriptive and diagnostic analytics for enhanced visibility into operational performance.
 - Level 2: Optimization Advisor – Offers predictive capabilities and actionable recommendations to improve production efficiency.
 - Level 3: Agentic Optimization – Enables autonomous, closed-loop optimization through agentic AI, reducing decision latency from hours to minutes.
- Physics Informed AI** - This capability integrates physics-based reservoir and production models with AI-driven analytics to ensure physically consistent, high-fidelity insights at accelerated speeds. By combining engineering rigor with machine learning adaptability, it supports rapid scenario evaluation, predictive modeling, and optimization of production strategies. The result is improved decision accuracy, reduced latency, and enhanced operational resilience across complex upstream environments.

The framework hosts multiple data driven and AI/ML applications such as Smart Reservoir Management (SRM), AI-based Voidage Replacement Ratio (Ai-VRR) forecasting, and Reservoir Pressure Mapping (S-PRES), among others.

Futuristic and Scalable AI Innovation

The AiPSO architecture is designed as a forward-looking platform that not only addresses current production optimization challenges but also serves as an incubator for emerging AI technologies. Its modular and scalable design enables rapid prototyping, validation, and deployment of next-generation AI solutions, ensuring continuous innovation and long-term digital resilience in upstream operations ([figure-2](#)).

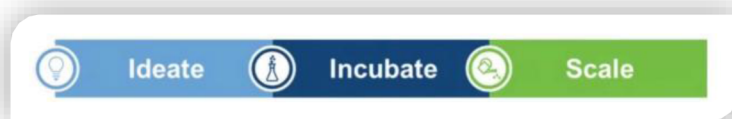


Figure 2—Innovation Lifecycle: From Ideation to Scale

AiPSO's architecture was designed for scale, trust, and adaptability, bridging legacy supervisory systems to intelligent orchestration across the upstream value chain. The platform comprises five core layers:

- Data Integration Layer
- Modelling Layer
- Workflow layer
- Agentic layer

the Data Integration Layer is responsible for ingesting, transforming, and harmonizing disparate data from IT, OT, and ET sources, laying the groundwork for a unified data environment. It ingests structured and unstructured data from SCADA systems, DCS, historians, maintenance logs, production forecasts, and simulation models, with streaming pipelines cleaning, tagging, and organizing these inputs in near real-time. This layer employs robust data governance frameworks to ensure data quality, security, and compliance, thereby establishing a reliable foundation for subsequent analytical and operational processes [8]. Data sources selected for the project are OsiSoft PI, and various engineering databases, enabling comprehensive data acquisition. To facilitate this data acquisition, the workflow leverages dedicated data extractors, which are specialized components designed to connect directly with source systems and push data into the data platform. For this project, pre-built extractors are utilized. These extractors are configured to stream or batch-extract data with minimal transformation, preserving the raw data fidelity. This approach ensures a continuous and reliable flow of data from its native environment into the central data platform.

The DataOps & Knowledge Graph Layer is a dynamic and semantically rich representation of the operational environment. Moving beyond traditional relational databases, which often create rigid and siloed data structures, the knowledge graph models the entire industrial ecosystem—including wells, networks, equipment, and analytical models—as a network of interconnected entities and relationships. This structure provides a unified, contextualized view of all data, which is crucial for breaking down information silos between IT and OT systems. This semantic layer acts as an intelligent data fabric, capturing not just the data itself, but the explicit meaning and context that are often lost in conventional data storage. By representing complex industrial relationships in a flexible, machine-readable format, the knowledge graph empowers both human experts and AI systems to perform complex queries and infer new insights. For example, it can trace the complete lifecycle of data, from sensor readings to analytical model outputs, providing a clear

lineage that is essential for data governance and explainable AI. Furthermore, the knowledge graph is the foundational component for leveraging generative AI [9]. It provides a trusted, verified, and contextualized data source that Large Language Models (LLMs) can use through Retrieval Augmented Generation (RAG). This ensures that AI-driven applications, from automated reporting to intelligent digital assistants, are grounded in the realities of the industrial environment, leading to more accurate and reliable decision-making processes. In essence, the knowledge graph transforms raw, disparate data into structured, actionable knowledge, enabling intelligent agents to access and interpret information far beyond simple data retrieval.

The workflow layer consists in an orchestration of multiple domain -specific workflows, designed to automate complex engineering and operational tasks across the upstream value chain. Each workflow integrates data from the knowledge graph, executes advanced analytical models, and generates insights or recommendations, ensuring a seamless flow from data acquisition to actionable intelligence [10]. These workflows are built upon a robust analytical framework, leveraging machine learning algorithms for predictive modeling, anomaly detection, and optimization, thereby transforming raw data into refined, domain-specific insights.

Finally, the Agentic Layer, featuring generative agents powered by LLMs and grounded in the knowledge graph, acts as an intelligent copilot. These agents retrieve contextualized data, invoke relevant domain workflows, interpret results, and present interactive outputs to users.

Smart Engineering Workflows

Engineering workflows form the operational backbone of AiPSO. These workflows have been selected, prioritized, and digitized to ensure they are aligned with the business drivers of increased uptime, reduced OPEX, and optimized production. By transforming manual, expertise-driven tasks into automated and intelligent processes, AiPSO unlocks both efficiency and scalability.

Over 25 digital workflows have been deployed across 8 oil fields in UAE and consist of a range of integrated smart engineering and AI/ML applications, from smart surveillance through real-time production optimization. These leverage data from the knowledge graph and integrate seamlessly with process simulators such as Prosper (for oil and gas wells) and GAP (for surface networks) [11]. As well, these are connected to the agentic layer to provide conversational intelligence and actionable recommendations. Workflows were grouped in four main categories to streamline implementation and maximize impact across different operational facets. Figure 3 illustrates a range of interactive visualizations that enable users to engage seamlessly with the smart engineering workflows



Figure 3—Representative AiPSO visualizations

● Surveillance and Diagnostics

- Artificial Lift Diagnostics: Integrated real-time sensor data with lift performance curves and digital well models to detect ESP instability, gas interference, or underperforming wells. Recommendations are generated automatically with decision confidence levels. This diagnostic capability leverages augmented AI to combine machine learning with deep domain expertise, enhancing decision-making in real-time well surveillance activities [12] [13]. This proactive

approach minimizes downtime and optimizes artificial lift performance by identifying potential issues before they escalate.

- Multi-Phase-Flow-Meter (MPFM) Validation: Compares MPFM readings against physics-based Nodal Analysis to identify discrepancies and suggest recalibration or flag potential measurement errors [14], [15] [13]. This ensures measurement integrity, a critical aspect for accurate production allocation and reservoir management decisions. Drift is detected using model-based thresholds and ML anomaly detection, providing a robust mechanism for maintaining measurement accuracy.

- **Production Optimization Workflows**

- Choke Optimization: Identifies suboptimal choke settings based on production targets and network constraints. Suggested setpoints are validated against model simulations. This ensures that adjustments maximize output while respecting operational limits and preventing detrimental flow dynamics.
- Injection Optimization: Uses surveillance and reservoir coupling models to assess sweep efficiency and recommends optimal injection allocation across multiple wells or clusters. This optimizes reservoir pressure maintenance and maximizes hydrocarbon recovery over the field's lifespan.
- Flow assurance: Monitoring of water chemistry and seawater injection parameters for scale management [16]. This workflow ensures that chemical treatments are proactively managed to prevent scale buildup, maintaining optimal flow rates and minimizing infrastructure damage.

- **Network and Facility Management**

- Network Surveillance: Continuously compares network simulation outputs with actual field conditions. Highlights pressure bottlenecks, compressor inefficiencies, and backpressure propagation.
- Facility Balancing: Ensures separator performance and facility utilization are aligned with production plans, adjusting flows where necessary. These capabilities streamline operational oversight, ensuring that facilities operate at peak efficiency and contribute optimally to overall production targets.

- **Event Detection and Recommendation**

- Exception-Based Management: All workflows operate on an exception basis, with alerts triggered by deviations. Each alert is automatically linked to diagnostics or optimization paths, enabling rapid triage.
- Value Attribution: Each workflow tracks value—e.g., BOE gained, downtime avoided, or optimization uplift. Dashboards roll up KPIs by asset and workflow, feeding into quarterly reporting. This holistic approach not only enhances operational efficiency but also provides clear, quantifiable insights into the tangible benefits derived from the intelligent automation of upstream processes.

AiPSO Agentic AI Architecture

The AiPSO Agentic AI Architecture leverages large language models to enhance human-machine collaboration in oilfield operations, moving beyond traditional automation by embedding sophisticated reasoning capabilities. This framework integrates autonomous decision-making agents that can interpret complex operational data, simulate various scenarios, and recommend actionable strategies, thereby transforming static workflows into dynamic, intelligent systems [17].

The Agentic AI in AiPSO is structured around a sophisticated three-layer agent architecture designed for complex industrial problem-solving. This modular design ensures that tasks are broken down and handled by specialized components, leading to more robust and interpretable outcomes. The core reasoning engine for these agents is a Large Language Model (LLM) that has been fine-tuned on extensive corpora of oilfield terminology, operational manuals, and Company procedures. This allows the agents to understand and communicate using either natural language or structured prompts while maintaining context across multi-turn dialogues.

The architecture consists of the following layers:

1. **Planner:** This top-level agent acts as the semantic orchestrator. When it receives a user query or detects an anomaly from monitoring systems, the Planner assesses the goal and formulates a high-level, multi-step plan. It deconstructs complex problems into a sequence of smaller, manageable tasks to be passed down to the other layers.
2. **Retriever:** Functioning as the information-gathering arm, the Retriever executes tasks defined by the Planner that involve data acquisition. It queries the industrial knowledge graph to fetch relevant context, such as equipment hierarchies, historical performance data, maintenance logs, active simulation inputs, or outputs from previous workflows. Its primary role is to supply the Executor with all the necessary, context-rich information to perform an action.
3. **Executor:** This is the action-oriented layer. The Executor receives data from the Retriever and invokes external tools, such as running physics-based or surrogate model simulations. After execution, it processes the outputs, synthesizes the results, and generates clear, context-rich explanations and recommendations for the end user.

The agentic framework is designed to address high-value operational challenges. Its capabilities are best illustrated through its primary use cases:

- **Root Cause Analysis:** When a well's or equipment performance deviates from the expected behaviour, an agent can autonomously initiate a diagnostic workflow. It retrieves real-time sensor data, historical production trends, maintenance records, and known failure patterns from the knowledge graph. By correlating this information, the agent can isolate probable root causes—such as downhole equipment failure, reservoir pressure changes, or surface facility constraints—and present a prioritized list of hypotheses to the engineer. This advanced analytical capability significantly reduces the time to diagnosis, enabling swifter corrective actions. Furthermore, thanks to the knowledge graph, the system can leverage its understanding of interconnected systems to identify cascading failures or multi-point anomalies that might elude human operators or simpler rule-based systems [18].
- **Proactive Scenario Simulation:** The framework empowers engineers to conduct rapid "what-if" analyses. An engineer can pose a question in natural language, such as: "What is the projected impact on the field's total gas-oil ratio (GOR) if water injection in Zone A increases by 10%?" The agent interprets the query, retrieves the appropriate surrogate model and baseline data, runs multiple predictive scenarios, and returns a comparative analysis, enabling data-driven operational planning.
- **Multi-Asset Optimization:** Operating beyond the single-well level, agents can perform field-wide optimization. By analyzing production deltas and constraints across a network of wells and facilities, an agent can recommend coordinated adjustments—like modifying choke sizes, reallocating gas lift resources, or altering injection strategies—to meet holistic targets, such as maximizing oil production while adhering to facility limits.

This approach represents a significant evolution from reactive problem-solving to proactive, predictive management of complex industrial systems [19].

For an AI system to be adopted in a critical industrial setting, it must be trustworthy and transparent. The AiPSO framework is built on a human-in-the-loop or "copilot" philosophy, where agents augment, rather than replace, human expertise. This collaborative approach is sustained by several key features designed to ensure explainability and user control [20], [21]. For engineers and field teams to accept each agent, it must present annotated outputs with clear references to the data, models, and reasoning steps used to reach a conclusion. The system should also flag uncertainties, such as recommendations outside established safe operating envelopes or based on low-confidence data, to ensure engineers are aware of potential risks. Additionally, each agent's actions must be logged for full traceability through audit trails, which are crucial for debugging, performance reviews, and continuous model improvement. This design fosters a dynamic partnership, allowing engineers to interrogate the system, override AI-driven decisions with their own rationale, and provide direct feedback for continuous learning and model refinement.

Deployment, Impact, and Strategic Implications

The AiPSO framework is being deployed across 18 distinct fields, delivering measurable improvements in operational performance and decision-making efficiency. Once fully implemented, the deployment is expected to yield an uplift in production capacity across these assets without necessitating additional capital expenditure (CAPEX). A key benefit is the significant reduction in decision latency—from hours to minutes—enabled by the agentic AI framework. Furthermore, the architecture is a highly scalable, with core optimization workflows being replicated across diverse assets with only minimal customization required.

The expected efficacy of the framework will be validated through several key performance indicators. AI agents are designed to provide optimization recommendations aligned with expert engineering analysis. Additionally, the system's ability to process cross-domain queries is anticipated to accelerate root cause identification for complex network anomalies. A critical success factor will be user adoption, which is being addressed through a focus on transparency and explainability in the agentic system's logic fostering trust and confidence among engineers. If successful, this initiative has the potential not only to enhance operational metrics but also to transform the collaborative paradigm between human operators and advanced intelligent systems in complex industrial environments, laying the foundation for a trust-based relationship crucial for sustained technological integration [22].

Strategically, the implementation of AiPSO represents a fundamental shift from passive, dashboard-based monitoring toward proactive, intelligent orchestration of field operations. As the framework continues to mature, it is expected to deliver several key benefits: greater operational agility, enabling field teams to respond to changing conditions with higher speed and confidence; a new model of human-machine collaboration, where engineers engage with AI agents as trusted "copilots" rather than opaque "black boxes"; and a significant advancement in the organization's digital maturity, laying the groundwork for future autonomous field operations.

The solution is positioned to deliver substantial value; however, several critical challenges must be addressed to achieve full-scale adoption and ensure long-term sustainability. Effective change management will be essential to drive user acceptance and seamless integration of agentic workflows among field personnel. The dynamic nature of reservoir and facility conditions will require ongoing model tuning and validation to maintain performance. Additionally, establishing a rigorous governance framework to oversee agent behavior, particularly in high-stakes workflows will be essential for ensuring safety and operational integrity.

Addressing these challenges will be central to the continued evolution and expansion of the AiPSO framework. Continuous validation and adaptive oversight will be paramount for maintaining the efficacy of agentic systems in dynamic operational environments [23]. Furthermore, the complexity introduced

by autonomous and intelligent systems necessitates governance structures that go beyond traditional compliance, emphasizing integrated, real-time supervision to manage inherent risks [24][25].

Conclusions

This paper presents a comprehensive overview of ADNOC's Agentic AI-driven Artificial Intelligence Production System Optimization (AiPSO) framework, highlighting its potential to transform upstream oilfield operations. The framework integrates domain-specific generative agents with a robust digital twin to enable autonomous diagnostics, scenario modeling, and real-time decision support—capabilities expected to deliver substantial production uplift and enhanced operational efficiency as implementation advances.

The ongoing deployment of AiPSO reflects a strategic shift from traditional reactive monitoring toward proactive, intelligent orchestration of field operations. This initiative is designed not only to optimize current workflows but also to establish a scalable and trustworthy foundation for future autonomous operations in the energy sector, redefining human-machine collaboration. Ultimately, AiPSO is positioned to deliver measurable production gains while accelerating ADNOC's digital transformation and strengthening its strategic positioning within the evolving upstream geoenergy landscape. ADNOC shall share measurable outcomes and success stories in upcoming industry forums and future events, showcasing the tangible value of agentic AI in driving operational excellence and advancing digital maturity across its assets.

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